Optimal Feature Identification for Machine Prediction of Wind-Wave Parameters at Wave Energy Converter Site

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***Abstract*— The hike in fossil-fuel prices and increasing environmental damage due to the subsequent Carbon Monoxide (CO) emission from burning fossil-fuel is becoming a major concern for every nation. The possibility of generating power from natural sources such as solar, wind, and sea waves is thus considered as an alternative. In the case of the sea waves, the kinetic energy of surface waves can be converted into single direction motion which runs a turbine to generate electricity. A Wave Energy Converter (WEC) is such an installation that converts the wave energy into electrical energy. In this study, we have conducted a literature investigation to identify the significant meteorological and wind-wave data parameters which determine wave-energy potential at a wave energy converter site and then identified optimal feature sets from buoy data for machine prediction of those identified parameters. The authors hope that by suggesting optimal feature sets, the outcomes of this study will help in improving the computational efficiency of machine learning models specially designed for wave parameter prediction at WEC sites.**

Keywords—wave-energy potential identification, significant wave height, wave energy period, wave peak period, feature selection, machine learning

I. INTRODUCTION

The need for effectively harnessing energy from non-fossil sources is crucial to decrease dependency on conventional fuel and to minimize the carbon footprint of a nation. The alternative sources for power generation are solar, wind, and marine energy such as sea waves, tides, and ocean thermal sources. These solutions are viable across larger geographical areas and have a high potential for commercialization. For such reasons, the topic of renewable energy has been eagerly investigated by government organizations, industry, and academic researchers.

Among all marine energy resources, wave-energy has an established global presence [12]. Additionally, the energy density of waves has been reported to be highest among all renewable energy resources [20]. However, its potential is yet to be fully exploited [16].

To harness wave-energy, a wave energy converter (WEC) device is required. The wave energy converter generally makes use of sea surface waves or wind-wave [14]. The basic principle of such a device is to converts the kinetic energy of the wave into a single directional motion which runs the turbine to generate electricity.

Due to the dynamic nature of wind-wave, it is important to understand its characteristics to estimate wave-energy potential for a specific location [14]. For example, in shallow waters (which is less than 100 meters deep), local bathymetry conditions also play an important role [12]. For these reasons, careful assessment and prediction of wave-energy resource at a candidate WEC site is important [5].

It is common to apply different techniques to evaluate and predict the potential of a candidate wave-energy site. These techniques can be divided into the following categories: (i) statistical modeling (ii) numerical modeling and (iii) machine learning methods. Among all these techniques, numerical wave modeling is dominantly used to evaluate wave-energy resources [3] [18]. These models are based on the energy balance equation [6]. Three generations of numerical wave modeling have been proposed [27] and currently, 3rd generation numerical wave models are used to assess wave-energy resources by modeling and predicting sea wave parameters and sea state [3] [5]. Examples of such models are Wave Model (WAM), Simulating WAves Nearshore (SWAN), and WAVEWATCH-III (WW3). Among these models, SWAN is predominantly used in coastal areas where the shoreline WEC facility is generally situated. The functional requirements of the SWAN model are wave boundary conditions, digital bathymetry, and a user-created computational grid. Based on this information, the model computes the transformation of the surface wave [10].

Despite its high accuracy and reliable results, a numerical wave model generally requires high computational power, involve complex mathematics, and need *in-situ* environmental and bathymetric data [10] [26]. Thus a swift and less complex approach is highly desirable.

In recent years, data-driven machine learning (ML) models have emerged as a surrogate to numerical wave models [11]. They have been successfully used to predict the wave-energy potential for a wave energy converter (WEC) facility [13]. These machine learning models are trained, tested, and validated using a data set. To realize the underlying function, the ML model analyzes the patterns in the data. In comparison to numerical models, machine learning models are easy to assemble and for site-specific wind-wave related prediction problems, they have outperformed numerical and statistical models [8] [9] [11].

It has been reported in the literature that real-time to short-horizon forecast (between 0 min to 6 hours) of ocean wave-energy is significant for WEC site, not only for its operational safety, optimal power generation but for its commercial viability as well [16]. Since wave parameters are important factors to characterize wave-energy [13] thus prediction of these parameters is important for wave-energy estimation.

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As discussed previously, numerical wave modeling has its inherited issues so as an alternative, machine learning models can be used to forecast these wave parameters. However, if not selected carefully, a tremendous amount of features in different wave data sets [1] can increase the computational cost of the machine learning model thus losing its edge over numerical wave modeling. Therefore, in this paper, we have focused on the identification of an optimal feature set for wave parameter prediction using the ML model for the WEC site.

The structure of the paper is as follows. Section II discusses the problem background and set the foundations for research questions formulation in section III. In section IV we have identified the answers to our research questions from literature and in section V we have suggested the optimal feature set for wave parameter prediction using ML model for WEC site and identified the future work.

II. Problem Formulation

As elaborated in section I, real-time to short-horizon prediction of wind-wave parameters are important for determining ocean energy forecast [16]. It not only improves the operational efficiency of the WEC facility but also safeguards the equipment from any potential damage due to prevailing or near-future rough sea conditions. Additionally, based on the wave condition prediction, certain safety measures can be taken to ensure the wellbeing of the individuals working on the site.

For a wave machine prediction problem, different sources such as wave buoy, radar, and satellite data sets have been proposed in the literature. The most commonly used data set is the wave buoy data set which is acquired from a strategically moored sea surface buoy. An extensive amount of time-series weather and wave data such as air temperature, humidity, precipitation, barometric pressure, wind speed, wind direction, water temperature, and wave parameters can be acquired from a buoy data set [1]. Moreover, the sea buoy data set has also been identified as a reliable source for wave-energy potential assessment studies [17].

Since feature selection is considered as one of the important steps in machine learning, hence from this plethora of meteorological and wind-wave feature set, the selection of optimal features set for ML model training is important. These selected features identify a subset, from original features, which provides the most significant information for the training of a machine learning model. Such an optimal feature set can significantly decrease the computational complexity of a prediction model, increases the prediction accuracy and generalization ability, and reduces the risk of overfitting [7] [2].

Recently, the selection of optimal sub-set of features from the buoy data set, for wind-wave parameter prediction has been identified as a challenge by the researchers [4]. To solve this issue, in section III, This study has formulated two research questions and their objectives to identify: (i) meteorological and wind-wave parameters relevant to WEC site, and (ii) optimal feature set to predict the identified parameters using machine learning model.

III. Research Questions

As discussed in section II, the diversity of wave buoy data set makes the identification of the optimal feature set for wind-wave parameters prediction problem a challenging task. For this reason, the study has formulated the following questions and their objectives to investigate the issue.

RQ1: Which meteorological and wave parameters are important to calculate wave-energy potential at a candidate wave energy convertor site?

RO1: To identify the significant meteorological and wind-wave data parameters which determine wave-energy potential at a wave energy converter site.

RQ2: Among different buoy data set features, which features plays a pivotal role in machine prediction of parameters identified in research question number 1?

RO2: To identify a sub-set of buoy data set features for optimal training of machine prediction of parameters identified in research question number 1.

IV. Investigation

This section attempts to find answers to aforementioned research questions. It is realized that, to answer RQ1, first, we need to understand the fundamentals of wind-wave generation and its significance for wave-energy calculation at the WEC site. We then examine the wave-energy assessment studies, to find an answer to RQ1. Later in the section, answer to RQ2 are investigated.

A. Wind-wave and its significance in wave power generation

It has been reported in the literature that power generation from sea waves requires careful selection of the location [5] and most of the WEC facilities harness power from wind-waves [14]. A wind-wave is a result of windblown over a large area of the sea [27]. During this process, wind power is transferred to the surface of the sea and surface waves gradually start to develop. These waves can carry the stored energy to a significant distance [19]. During this process, very little energy is wasted. This stored wave-energy is then harnessed at a WEC site. Thus, not only the prevailing wave but weather conditions as well, play a significant role in determining the power generation potential of a site.

The existing wind-wave conditions define the sea state [27], which makes up the wave climate. A wave climate is defined as a site-specific long-term statistical characteristic of the waves [15] so it is obvious that for a specific location, the more consistent the wave climate is, the more power generation potential it has. However, we need to understand that the sea state is eventually a result of metrological conditions and these conditions vary across the time so temporal variations in sea-state are present. Thus we are dealing with the dynamic conditions of the sea which ultimately defines the energy potential of a site.

B. Important considerations for wind-wave energy converter site

As explained in the previous sub-section, the regional temporal characteristics of sea sate are one of the important factors in consideration of WEC facility installation. Directional variation of wave climate has been identified as another factor for assessing a power generation site. The increase in the directional variation of a wave climate has been attributed to less power generation. Additionally, spectral characteristics of wave climate are interrelated with the wave spectrum. Since the efficiency of wave energy converter is dependent on a wave frequency thus a specific wave frequency may generate more power as compared to other

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frequencies. For this reason, it is important to evaluate the wave spectral characteristics of the WEC site as well [14].

C. *Significant parameters for wave-energy potential assessment*

Barstow et al. [12], has identified buoy data set as one of the main data source for energy source site assessment. To cover different seasonal and ocean-atmosphere climate cycles, a data set covering multiple years is suggested by the authors. For power assessment of a specific site, wind-wave/swell parameters such as significant wave height (H_s), mean wave direction (θ_m), peak period (T_p), mean wave energy period (T_e), and wind parameters such as wind speed (U_{10}) and wind direction (θ_w) were considered in their study.

The proposed probabilistic forecasting of wave-energy by Pinson et al. [16] was based on wave energy flux equation which is a stochastic process and is defined as (1).

$$Y_t := \frac{g^2\rho}{64\pi} H_t^2 T_t, \qquad Y_t \geq 0 \tag{1}$$

Here g represents gravitational acceleration constant, ρ denotes seawater density, H_t represents significant wave height and T_t is mean wave period. The study acquired meteorological data from WAM, a 3rd generation numerical wave model, and wave parameters from the buoy data set. However, contrary to [12] they opted to choose significant wave height and wave period as an input feature for their model.

For coastal wave-energy assessment, Veigas et al [5] used 3rd generation numerical model SWAN. The numerical wave modeling was based on wave parameters acquired from offshore buoy and bathymetry information of the coastal area. The buoy data had an interval of 1 h. In this study, it is noticed that significant wave height was again chosen as one of the parameters in addition to, energy period (T_e) and mean wave direction (θ_m).

Vannucchi and Cappietti [22] identified candidate spots for WEC site in the Mediterranean Sea by applying: (i) wave energy flux equation (1) using significant wave height (H_s) and energy period (T_e) for off-shore assessment of wave-energy, and (ii) Spectral Wave (SW) numerical model using significant wave height (H_s), peak period (T_p), mean wave direction (θ_m) and its spreading factor for near-shore assessment. It is important to notice that the parameter H_s is commonly used in both off-shore and coastal studies. Additionally, the selection of T_e and T_p , as per location, is an interesting find.

Guillou [31] estimated wave-energy flux using significant wave height (H_s), wave energy period (T_e), and peak wave period (T_p). The energy-period formulation was computed using (1), however, due to scarcity of T_e information in hindcast archives, it was calculated from available T_p data by applying a calibration coefficient (α) such that

$$T_e = \alpha T_p \tag{2}$$

The wave-energy flux equation is then defined as (3).

$$Y_t := \frac{g^2\rho}{64\pi} H_t^2 \alpha T_p \tag{3}$$

They reported good wave power density estimations in deep waters by using the energy-period formulation, as reported by [22] as well. However, peak-period based estimations were reported more than available energy flux. This overestimation was then minimized by a refined distribution of α against H_s and T_p.

Based on these studies we can infer that

1) The parameter, significant wave height (H_s) is commonly used in wave-energy assessment studies [12] [16] [5] [22] [31].

2) Energy period (T_e) based estimations are suitable for off-shore wave-energy estimations [22] [31] while near-shore estimations are influenced by peak period (T_p) [22].

3) Other parameters have also been used for wave-energy assessment, however the study [31] suggest that parameters such as H_s, T_e, and T_p can be used to estimate wave-energy potential at off-shore and near-shore sites.

4) It was observed that for coastal power assessment, information about local bathymetry is also important [5] [12]. However, this finding is more relevant for the numerical model-based assessment of wave-energy.

TABLE I. DATA PARAMETERS OF WAVE-ENERGY POTENTIAL ASSESSMENT STUDIES

Study	Data Source	Period Covered	Temporal Resolution	Parameters Considered
Barstow et al. [12]	Sea Buoy, Satellite Altimeter, Wave Model	10 years	6 hours	Significant Wave Height, Mean Wave Direction, Peak Period, Mean Wave Energy Period, Wind Speed, Wind Direction
Pinson et al. [16]	Sea Buoy,Wave Model	13 months	3 hours	Significant Wave Height, Mean Wave Period
Veigas et al. [5]	Sea Buoy, Bathymetry	15 years	1 hour	Bathymetry Information, Significant Wave Height, Energy Period, Mean Wave Direction
Vannucchi and Cappietti [22]	Wave Model	4.5 years	3 hours	Significant Wave Height , Wave Period, Peak Period, Mean Wave Direction, Wave Spreading Factor
Guillou [31]	Sea Buoy	9 to 22 years	1 hour	Significant Wave Height, Wave Energy Period, Peak Wave Period

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By analyzing Table I and our literature findings, we can identify significant wave height (H_s), wave energy period (T_e) , and wave peak period (T_p) as the important parameters for determining wave-energy potential at a WEC site. Based on this conclusion against our investigations for RQ1, the study now attempts to find an answer to RQ2 by identifying a sub-set of wave features for optimal training of machine prediction of $H_s, T_e,$ and T_p parameters.

D. *Optimal feature set for machine prediction of significant wave height, wave energy period and wave peak period*

The buoy data set consists of a large number of parameters [1], and using all these parameters for the machine prediction model is not a cost-effective solution. For this reason, the feature sets reported in the literature to predict $H_s, T_e,$ and T_p will be investigated in the following text.

Among 10 different parameters, wind speed (U), gust speed, air temperature (T_a), atmospheric pressure, and water temperature were recognized by Cornejo-Bueno et al. [23] as most affective training features for H_s prediction for wind-sea. However, Akbarifard and Radmanesh [30] have identified significant wave height (H_s) and wind speed (U)as features of great interest to the same problem.

Abhigna et al. [24] predicted H_s using wind speed (U) , gust, and direction (θ) parameters. Hashim et al. [25] has also identified wind speed (U) and wind direction (θ) as an important feature for H_s prediction , however, they suggested the addition of sea surface air temperature(T_a) to improve the accuracy. Interestingly, in a recent study [21], the historical values of significant wave height were solely used for H_s prediction.

For wave energy period prediction, Barker and Murphy [29] have identified H_s, first wave period (T_1), T_p and peak direction (D_p) as the optimal feature set for ML prediction. Among all these parameters, T_p was identified as the most influential parameter for the prediction of T_e.

A study for wave peak period prediction has identified wind speed (U) as the affective parameter when using wind duration (tr), H_s and $T_p$as other input features for ML model [28]. The wind speed was also identified by Mahjoobi et al. [26] as an affective parameter to predict T_p , along with wind direction (θ), fetch length and wind duration.

From given evidence, it can be concluded that

1) Significant wave height prediction can be optimally done by either using historical H_s data [21] or wave speed, direction and air temperature data [25].

2) Wave energy period prediction is affected by T_p.

3) Wave peak prediction is influenced by wind speed and wind duration [28] [26]. Thus they should be selected from feature list.

4) Half to one hour temporal resolution of feature set is generally sufficient for wave parameter prediction [21][30][23][24][25][29][28][26].

V. Conclusion and Future Work

For the prediction of wind-wave parameters at wave energy converter site, a literature investigation study to identify optimal features for machine learning model was conducted. The study attempted to identify: (i) the significant meteorological and wind-wave data parameters which determine wave-energy potential at a wave energy converter site, and (ii) an optimal sub-set of buoy data set features for training of machine prediction model of those wind-wave parameters.

Through the literature investigation, it was found that significant wave height(H_s), wave energy period (T_e) , and wave peak period (T_p) are significant parameters to determine wave-energy potential. Later, optimal feature sets from literature were also identified for machine prediction of $H_s, T_e,$ and T_p.

For future work, a more detailed study can be carried out to propose a single optimal features set for prediction of all three identified parameters (i.e. $H_s, T_e,$ and T_p).

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